ence in the amount of carbon, and naturally in the hardness of the tire, in the tensile strength and natural elongation between the different makes. They run from about .055 carbon to .072 carbon. I speak of car wheel tires, and take it for granted that the same variation exists between the tires of locomotive wheels. If the tire is of the higher carbon it would naturally have less elongation and less liability to roll out, and the rolling out and stretching would undoubtedly be very much less than in the case of the softer tires. The question, therefore, arises, whether it would be better to raise the carbon points in the tires, so that you will have a harder tire and one less liable to this rolling-out action, than when you have the softer tire such as we have been using. I want to ask whether the committee made any measurements to find out what the stress on the tire is, after it is shrunk on. Of course, there must be some stretching to the tire, and I ask whether that tensile stress is ordinarily calculated to take up the whole of the shrinkage, and whether measurements have been taken to show what the depression of the center is by which the actual tensile stress on the tire, when finished, is somewhat less than it otherwise would be.

Mr. S. T. Park (Mich. Cent.): Two makes of tires were used in getting these dimensions, and both showed the rolling action. Although we did not make a chemical analysis we made a physical test of one of the tires, and it gave an ultimate strength of something over 100,000 lbs.

Mr. G. W. Wildin (Erie): We are abandoning retaining rings altogether, up to 76-in. driving wheels.

Mr. W. G. Menzel (Wis. Cent.): We don't need a retaining ring when we have tires 3½ in.; only when we get them down as low as 2½ in. It is a question whether it is best to put retaining rings on in order to get another ¼ or ½ in. wear on the tire. We don't need the retaining rings until a tire gets down to 2 in. Would it pay to put it on to get another ¼ in.? I don't think it would.

Mr. F. Slater (Chic. & N.-W.): With the projection used in the Pennsylvania form it would be difficult to shim the tire; but if the tires are fitted with an inside retaining ring, they could not shift, and an engine developing loose tires on the road could be brought to her home station and it would save the expense of sending men out on the line.

Mr. H. H. Vaughan (C. P. R.): We are using the lip shown in Fig. 6 on our tires and shimming them from the inside where necessary, and I don't believe we have any very serious difficulty. It accomplishes the same purpose as the retaining ring, and is an element of safety.

In regard to the design of the steel wheel, referring to Fig. 4: The center line is shown ½ in. inside of the spoke, and the center line of the tire is about another ½ in. on the outside. In Fig. 2, showing the spokes, you will see that the center of the thrust of the tire, tending to collapse the wheel center, is very near the inside end of the spoke which the committee has made considerably thicker than the outer end. We have gone a step farther than that. We have put the bulk of our metal in the spoke right under the line of the thrust of the tire, making a pear-shaped spoke and reduced the section very considerably on the outside. We started to make practically a T-shaped spoke and came to the pear-shaped spoke. The wheel-makers say the wheel is a very easy one to mould, and it appears to me correct, because it puts the bulk of the metal right under the thrust of the tires. I think that section is preferable to the one suggested by the committee.

I don't believe the rim section suggested by the committee is correct, either. We have a considerable number of engines with cast steel wheel centers, with a rim very similar to this, although perhaps not quite so heavy, and as the tires were thinner we found considerable trouble from loosening, and also we have had trouble from breakages. I charge that entirely to the bending of the rim, with a thin tire, between the spokes. In the old designs of cast iron wheel it was common to have a rim 3 to 3½ in. deep. That rim had plenty of strength. It would support a thinner tire, and it had strength to do it. Our rims now are all made with practically the same section as the old cast iron centers with the deep U-shaped rim. While it takes no more metal than the proposed rim, it gives a very large additional amount of strength to support a thinner tire. I believe that is better than the section shown here.

We have been making experiments to determine the amount the wheel center collapses when a tire is shrunk on. For instance, you apply a tire with a shrinkage of .001 in. to an inch. We happened to strike a design of tender wheel centers that were over size. We would normally have given them .030 in. shrinkage. They were .030 in. over size. We were astonished on measuring the tire, after being shrunk on, to find it had come down to within two-one-thousandths of its cold diameter. The wheel center had given away entirely. That type of wheel center is one on which we have considerable difficulty with loose tires, and I think that accounts for it. When you put on .020, or .030, or .040 shrinkage, if you measure the diameter after it is put on, you will find the centers have collapsed. Driving wheel centers collapse under .025 to .040. The tendency in the pear-shaped spoke with the metal in the direct line of the thrust of the tire is to hold the wheel out and not give away as much as the tire with the spoke with the section shown by the committee.

Another point is that where rims are cut they should be fitted with cast-iron spacers. We struck some engines turned out a year ago, where they were fitted with soft metal spacers. They took away the entire resistance of the rim to hold out against the shrinkage of the tire.

Mr. F. Slater (C. & N.-W.): We found no difficulty in shimming the tire with the lip without placing a jack under the engine or removing a rod. We merely put a jack under the crank-pin to take enough of the weight off to free the tire from the rail. It does not take much pressure. There is little strain on the pin. It merely raises the wheel off the rail enough to take the weight off the tire. Then a tire heater is put aronnd the tire, and in a very few minutes it is warm enough to put tin all around the wheel center, and the engine can then go out. It saves a lot of roundhouse work. We shim a tire that way out on the road or in the house. For work on the line we have a small portable tire heater and send two men out. They use the air from the locomotive, and in a very short time the engine can go on with its train.

Mr. P. Maher (T., St. L. & W.): I am experimenting now with a retaining ring the full diameter of the wheel, to be put on in sections. It acts more as a clamp. That leaves a place between each clamp for shimming.

Mr. D. J. Durrell (P. L. W. P.): The action of the brakes probably has the greatest tendency toward loosening the tires, especially upon thin tires. We have recently made some investigations as to the thickness tires should be run on Atlantic type engines before being taken out of service, and we have found that 2 in. should be the limit.

President Ball: What disposition do you wish to make of this report, gentlemen?

Mr. C. A. Seley (C., R. I. & P.): We have the matter before us now in the form of a tentative report from the committee, and I would make a motion to lay it over for one year before being submitted to letter ballot. (Carried.)

Flexible Staybolts.—The report was read by the Secretary. Mr. W. H. Newbury, of the motive power department of the Pennsylvania, was present to represent the chairman of the committee, Mr. R. M. Durborow.

Mr. H. H. Wickhorst (C., B. & Q.): In making these measurements of the movements of sheets, I would ask if the sheets were free from scale? Was the fire-box sheet new?

Mr. Newbury: No, not a new sheet.

Mr. Wickhorst: It strikes me that probably the main reason for large movements of such sheets is the forming of scale, or mud or other insulating material, on the fire-box sheet and staybolt. If the sheets and staybolts are free from scale, the temperature cannot be very much different from the temperature of the water. I made some temperature tests which indicated with a hot fire that a ½-in. sheet cannot be raised over the temperature of the water more than 40 or 50 deg. F.; but allow ⅛ in. of scale to form, and the temperature of the metal is going to rise above the temperature of the water by several hundred degrees. In fact, with a hot fire and ⅛ in. scale on the staybolts and sheets, the sheets can actually attain a low red heat. As the relative movement of the sheets is probably primarily a function of the difference of temperature more than any one thing, it will be seen that if the inside sheet is allowed to become materially raised in temperature, the movement is going to be greater, and I rather imagine that if the test had been made with scaled boiler sheets, that movements would have been found, as, for instance, by opening the fire-door, after a good hot fire. With the hot fire and the scaled sheet, with the temperature of the water, say 400 deg. or lower, the temperature of the fire-box sheet would easily be raised to 600, 800 or 1,000 deg., whereas if the sheet is perfectly clean the temperature of the fire-box sheet probably would not have been raised to over 450 deg.

Mr. G. R. Henderson: I do not notice anything in the report about the cost of staybolts, and that is an important item, particularly if you have side sheets, which only last a year or two in service. We can hardly afford to put in very extensive staybolts in those sheets.

Mr. J. F. Walsh (C. & O.): We commenced using the flexible staybolts about three years ago on our high-pressure engines. We had a class of consolidation engines, with long fire-boxes, carrying 200 lbs. of steam, which troubled us greatly. As we commenced the construction of the wide fire-box engines, we put the staybolts in the outside rows in the side sheets, butt sheets and additional ones up in the upper corners, and putting as many flexible staybolts in the throat sheets as we could get in there. We probably have, all told, 15,000 or 16,000 flexible staybolts in 210 or 215 engines, and up to a few days ago we had not a failure of the flexible staybolts. However, two days ago I got a report of 22 broken flexible staybolts removed from one of our engines, which had been in service about 23 months. That is the first failure we had out of the 15,000.

Mr. A. E. Mitchell: I have had experience with the flexible staybolt of the type shown in Fig. 7, which is the only type that I have used. All of my experience has been on the Wooten type of boilers. We have introduced about 40,000 of these bolts in the last two years, and in that period we have not had a failure of that type of bolt.

sq. in. in area, with form as shown in Fig. 1, for the big end. Section at small end to have an area of from 5¾ to 6 sq. in., with form as shown in Fig. 2. These sections are taken at the base of the fillets uniting the spoke to the hub and rims, as shown in Fig. 3 at AA and BB. The metal in these sections is distributed so as to give the maximum amount of resistance to the shrinkage of the tires, and with this end in view the section is much heavier directly underneath the part of the tread which bears directly upon the rail head.

Rims.—Cast-steel driving wheel centers should be preferably

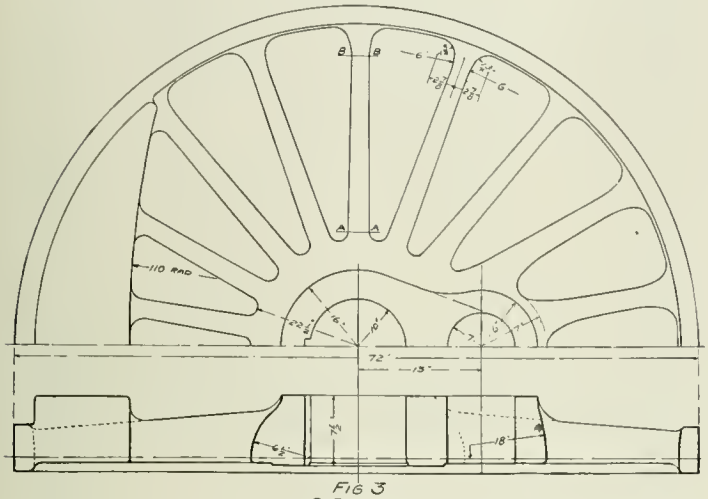


FIG 3
18 SPOKES

cast with the rims uncut and shrinkage slots omitted whenever steel foundries will guarantee satisfactory castings. The Pennsylvania Railroad has been obtaining wheel centers with uncut rims for the last two years or more. A number of cast-steel foundries are now ready to make wheel centers in this manner.

For wheel centers 60 in. and over, where the permissible total weight of the engine will permit, the rims should preferably be cast solid without cores, so as to obtain the maximum section and have full bearing of tires. The section in square inches should be approximately .45 of the sectional area of the tires when new; for instance, a 5½ by 3½ in. flanged tire has a sectional area of 20.2 sq. in. This would require a rim with 9 sq. in. of metal. A 3 by 5½ in. flanged tire has a sectional area of 17.5 sq. in., requiring a rim of approximately 7.9 sq. in. In our opinion, it is not necessary to consider the increased area of flat tires.

It is difficult to get sufficient counterbalance in centers smaller than 60 in. in diameter, so that it will be found very desirable to core out the rims to obtain the maximum lightness on the side next to the crank pin, and in some cases on the counterbalance side, in order to fill in with lead where necessary. The committee would recommend a rim section as shown in Fig. 4 for wheel centers without retaining rings. The tire is secured from having the center forced through it by a lip on the outside ⅜ in. wide and about ⅛ in. high, the tire being left rough at this point. The exact height of the lip therefore depends upon the amount of finish left on the interior of the tire.

Retaining Rings.—The committee does not recommend any particular form.

Shrinkage Slots.—It is becoming more and more the practice

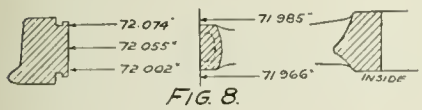
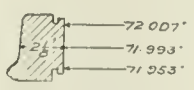


FIG. 8.



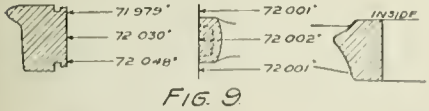


FIG. 9

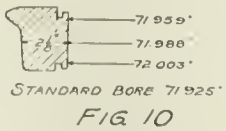


FIG. 10

of steel foundries to cast rims without shrinkage slots and several of them are now prepared to guarantee satisfactory results. The committee recommends the adoption of this practice whenever foundry methods will permit, as the rim of the wheel is stronger and better adapted to resist the strains to which it is subjected when the rim is left uncut. The extent to which this practice may be used depends entirely upon the attitude of steel manufacturers. It is well known that the shrinkage strains can be taken care of and that cast-steel wheel centers of all diameters required for locomotives have been made successfully in this manner for several years.

Hubs.—In order to prevent an excessive dish to the spokes the committee recommends that a standard distance of 55 in. between hubs be adopted. This permits of the use of a practically straight spoke without dish on the inside.

In boring out the tires and in turning off wheel centers, a wide-nosed tool should be used and care taken that the surface of the finishing cut is not left in ridges. It has been observed that when ridges are left they soon wear flat and decrease the original shrinkage.

Accurate measurements of tires after they have been in service for some time, especially when less than 2½ in. in thickness, show that a rolling out or stretching of the tire occurs, and for reasonably heavy centers these figures would account more for loose tires than any permanent set in the driving-wheel center.

In Fig. 8 is shown the exact caliper dimensions of the wheel centers and tires which were reported loose on Atlantic engines having about 104,000 lbs. weight on drivers. These tires had been previously shimmed, and contained liners between the tire and center .068 in. on the left and .050 in. on the right for the entire circumference.

Fig. 9 shows diameter of tires 2⅛ in. thick, which had been removed from a similar engine. These dimensions appear to confirm the statement that considerable rolling out and stretching occurs when the tire becomes reduced in thickness, and it is especially noticeable that this rolling action of the wheel is most apparent on the outside of the tire where the wear of the rail would be greatest.

The report is signed by F. J. Cole, J. E. Muhlfeld, A. S. Vogt, W. A. Nettleton.

SUBJECTS.

For Noon-Hour Topical Discussion, Convention of 1906.

1. Is not a boiler pressure of 185 lbs. better than 200 lbs. for locomotives?

Opened by Dr. W. F. M. Goss.

2. The necessity of proportioning brake pressure to wheel loads.

Opened by G. L. Fowler.

3. Is it practical to use the Prosser tool in roundhouse running repairs?

Opened by F. R. Cooper.

4. Grease vs. oil in driving box cellars.

Opened by F. H. Clark.

5. (a) Relation of roundhouse to shop and road.

(b) The best roundhouse organization for properly taking care of locomotives.

Opened by S. W. Miller.

6. Distortion of wheel centers and tires out of round due to heavy counterbalance on 100-ton engines.

Opened by George W. West.

7. To what extent should an engine be repaired in the main shop and what class of repairs could be made to advantage in the roundhouse?

Opened by C. A. Seley.

8. Relative advantage of the piston valve as compared with the slide valve.

Opened by E. A. Miller.

9. Is the Walschaert valve gear an improvement over the Stephenson link movement?

Opened by George W. West.

Committee Reports, 1907 Convention.

1. The comparative arrangement for coaling stations.

F. F. Hildreth, F. H. Clark, J. B. Barnes.

2. Submit a blank form for use at terminals to give history of the movement of every locomotive from the time it leaves its train until it takes its next train. Object: To secure closer co-operation between the mechanical and operating departments.

G. M. Basford, R. D. Smith, R. P. C. Sanderson.

3. What thickness of tires is the most practical for locomotive driving wheels—3 in., 3½ in. or 4 in. thick? Why?

W. T. Fitzgerald, John Howard, H. F. Bentley.

4. Results of different valve gears on locomotives.

C. A. Seley, A. S. Vogt, G. H. Haselton, Robert Quayle.

5. Proper allowance for fitting axles and crank pins to driving wheel centers and method of registering the pressure.

S. W. Miller, L. R. Pomeroy, S. M. Dolan.

6. Pay for shop and roundhouse labor and comparative merits of day work, piece work, bonus or other systems.

David Van Alstyne, J. F. Deems, W. R. McKeen, Jr.

7. The proper spacing of flues in high pressure boilers.

Charles E. Fuller, H. J. Small, F. J. Cole.

8. Supervision in larger locomotive repair shops.

G. W. Wildin, H. H. Vaughan, J. B. Kilpatrick.

9. Boiler shop area in square feet, per pit of erecting shop, including cranes, machine and tool equipment.

T. Rumney, A. C. Deverell, A. E. Manchester.

Individual Papers, 1907 Convention.

1. Locomotive failures, records and results of keeping them.

Paper by W. E. Dunham.

2. Superheating and compounding from a standpoint of locomotive economy.

Paper by T. A. Foque.

3. Shop cost systems and effect of shop schedule on output and cost of locomotive repairs.

Paper by A. Lovell.